BioKEA Laboratory Operations

BugPicker User Guide

Setup, automated plating, image review, taxonomy, QA labeling, and model maintenance

Guide build: October 12, 2026
Applies to: the software currently checked into this BugPicker folder

IMAGE NEEDED

`machine-overview.jpg`

BugPicker overview with sorting trays, destination plate, recovery plate, N1, Top camera, bottom camera, and HiRes camera identified.

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1. Purpose and Responsibilities

BugPicker scans sorting trays, identifies candidate specimens, picks with vacuum nozzle N1, places specimens into a 96-well plate, verifies nozzle and well images, and records images and metadata. Separate programs support hand-plated wells, taxonomy review, QA labeling, model evaluation, retraining, and candidate promotion.

> **Scope.** Routine operators should not change machine limits, camera definitions, G-code, model architecture, or script source.

The operator must confirm tray and plate placement, watch initial motion, answer review windows accurately, preserve specimen identity, and stop whenever physical motion does not match the displayed plan.

2. Safety and Emergency Response

IMAGE NEEDED

`emergency-controls.jpg`

Machine power and emergency controls.

> **Stop immediately** if a nozzle approaches hardware at an unsafe height, a tray is not seated, coordinates are clearly wrong, or a cable can snag.

1. Use the run Halt control when there is time to stop normally.
2. Use machine power or emergency stop for an imminent collision. Do not rely on a dialog.
3. After a crash, inspect N1, holders, cameras, plates, and loose specimens before motion.
4. Restart and home only after removing the obstruction. Watch homing continuously.

> **Homing is not proof of clearance.** Confirm N1 is at safe travel height before XY motion. Never leave the first movements unattended.

3. System Overview

Component	Role
Top/Lumen camera	Scans trays and captures whole-well QA images.
HiRes/Teslong camera	Captures detailed specimen and well images for taxonomy and records.
Bottom camera	Inspects N1 after picks and placements.
N1	Primary vacuum pick/place nozzle.
Sorting tray	Holds specimens; the automation profile uses size-dependent heights.
Destination plate	96-well plate. H12 is reserved as the negative control.
Recovery plate	Receives/wipes material remaining on N1.

User programs are under `Scripts → BugPicker → 00_Run_BugPicker`. Numbered and Python files are implementation components.

IMAGE NEEDED

`scripts-menu.png`

OpenPnP menu showing 00_Run_BugPicker.

4. Starting OpenPnP

- ☐ Clear the complete travel area.
- ☐ Seat trays, plates, and holders.
- ☐ Connect cameras and machine USB devices.
- ☐ Start OpenPnP and enable power.
- ☐ Home while watching the head and N1.
- ☐ Verify Top, bottom, and HiRes live feeds.
- ☐ Verify N1 can reach safe travel height.

IMAGE NEEDED

`openpnp-home.png`

OpenPnP after successful startup and homing.

Resolve any gray, frozen, or red-X camera before beginning. A disconnected HiRes camera may require closing OpenPnP, reconnecting USB, and restarting.

5. Preparing a Plating Run

1. Seat each sorting tray consistently; no edge should be raised.
2. Spread specimens to reduce overlap.
3. Seat destination and recovery plates in calibrated slots.
4. Record collection code, plate number, starting well, tray assignments, and classifier modes.
5. Open `RUN_BUGPICKER.js`.

IMAGE NEEDED

`tray-loaded.jpg`

Correctly seated automation tray with separated specimens.

Run Layout

The setup window defines active collection events, trays, plate numbers, starting wells, debris/taxonomy modes, and auto re-plate. Tray-to-event assignments protect sample identity.

IMAGE NEEDED

`run-layout.png`

Main run-layout window.

Option	Use
Train Debris Classifier	Presents uncertain targets and records operator feedback.
Auto-Debris Classifier	Uses the active model to include/exclude targets.
Train Taxonomy	Uses operator taxonomy review.
Auto Taxonomy	Uses the active taxonomy model where supported.
Auto re-plate	Continues while open wells and eligible targets remain, subject to no-progress and generation limits.

6. Calibration

> **Calibrate to seated hardware.** Tray height and tilt can change with seating.

The window controls tray bounds/profile, plate positions, HiRes-over-A1 and focus, and recovery position. Use jog and Record controls instead of estimating coordinates.

IMAGE NEEDED

`calibration-window.png`

Calibration window with tray, plate, HiRes A1, and recovery controls.

1. Select the correct slot and tray profile; use Automation tray for rigid automation holders.
2. Verify scan bounds with the Top camera.
3. Center HiRes over A1 and confirm the focus sweep is collision-free.
4. Verify Top-camera A1 and recovery A1 for automated plating.
5. Raise N1 before accepting calibration.

Automation-tray depth depends on specimen size and can descend progressively on repicks, never below the physical floor. Repeated initial-height failures can trigger adaptive rescanning; no-progress

and generation guards prevent endless loops.

7. Running BugPicker

1. Watch N1 rise before scanning.
2. Verify the first scan row is over the tray.
3. Review targets; exclude debris, unsafe overlaps, and unpickable items.
4. Watch the first complete pick and placement.
5. Monitor camera feeds, nozzle QA, well QA, and recovery.
6. Do not close OpenPnP while intentional child processes are active.

IMAGE NEEDED

`scan-review.png`

Target review showing included and excluded targets.

Decision Sequence

Bottom inspection confirms material after a pick. If none is detected, the well remains unattempted. After placement, the Top whole-well image is evaluated first. Occupied wells receive a HiRes image; apparently empty wells can be retried after bottom inspection. Material remaining on N1 enters recovery/cleaning.

After three recovery attempts, the current workflow continues rather than ending solely on a likely nozzle false positive. Images remain available for labeling.

8. Reviewing and Completing a Plate

The summary distinguishes attempts, successful placements, second-attempt successes, and wells needing review. Treat the physical plate as authoritative.

IMAGE NEEDED

`plate-review.png`

Completed-run plate review or filled plate map.

1. Review wells flagged potentially empty and correct occupancy.
2. Review the full plate map for unexpected gaps.
3. Confirm images and rows correspond to physical wells.
4. Continue auto re-plate only while valid specimens remain.
5. Review the completed-plate audit before taxonomy.

> H12 is the negative control and is not filled.

9. Hand Plating and Manual Imaging

Use `RUN_MANUAL_PLATE_IMAGING.js` for a plate filled wholly or partly by hand.

IMAGE NEEDED

`manual-imaging.png`

Manual setup and HiRes-over-A1 calibration.

1. Place the plate in the normal destination position.
2. Enter plate number, collection code, and first/last occupied wells.
3. For an existing BugPicker plate, keep **Do not replace wells already in the spreadsheet** selected. Recorded occupied wells are skipped, including around gaps.
4. Move HiRes to A1, center/focus, and record the position.
5. Start and watch the first movement.

> This mode does not detect occupancy. Every spreadsheet-empty well in the selected range is assumed hand-filled. Exclude physically empty wells.

Normal plate images, CSV, and XLSX are updated with `manual plating HiRes` confirmation.

10. Taxonomy Workflow

Use `RUN_TAXONOMY.js` for a completed plate without running the picker.

IMAGE NEEDED

`taxonomy-review.png`

Taxonomy review with identification, image consistency, and notes.

1. Confirm all images belong to the same well/specimen.
2. Review proposed taxonomy rather than accepting confidence alone.
3. Record image consistency accurately. Notes other than “Same specimen across images” enter the plate Notes field.
4. Complete/save review as directed.
5. Verify taxonomy outputs and spreadsheet before archiving.

11. QA and Model Maintenance

Use the candidate workflow: label, evaluate, train, audit, then promote after reviewing the recommendation. Training does not replace the active model automatically.

IMAGE NEEDED

`qa-label-review.png`

QA label review with available labels.

IMAGE NEEDED

`model-report.png`

Candidate report and promotion recommendation.

1. Collect representative empty and occupied nozzle/well images.
2. Run `RUN_QA_LABEL_REVIEW.js`; prioritize errors, low confidence, difficult lighting, and unreviewed images.
3. Run `RUN_QA_CV_EVALUATION.js` for cross-validation without changing the active model.
4. Run `RUN_QA_MODEL_TRAIN.js` to train and evaluate a candidate.
5. Run `RUN_QA_CANDIDATE_AUDIT.js` for nozzle safety and empty-clearance gates.
6. Promote only when supported by the report or a maintainer documents an override.
7. Restart the relevant workflow after promotion so it reloads the model.

Dataset size alone is insufficient. Nozzle QA needs truly empty examples spanning runs, rotations, lighting, and nozzle appearances that resemble live failures.

12. Program Directory

> `RUN_BUGPICKER.js`
>
> Main scan, review, pick/place, QA, recovery, replate, summary, and audit workflow.

`RUN_MANUAL_PLATE_IMAGING.js`

HiRes images and records for hand-plated wells.

`RUN_TAXONOMY.js`

Taxonomy classification/review for an existing plate.

`RUN_CAPTURE_EMPTY_NOZZLE_QA.js`

Captures confirmed-empty N1 images with rotation variation.

`RUN_CAPTURE_EMPTY_WELL_QA.js`

Captures confirmed-empty whole-well Lumen images.

`RUN_QA_LABEL_REVIEW.js`

Labels nozzle/well QA images, including prioritized difficult samples.

`RUN_QA_CV_EVALUATION.js`

Cross-validates QA behavior without promotion.

`RUN_QA_MODEL_TRAIN.js`

Trains a QA candidate and reports a recommendation.

`RUN_QA_CANDIDATE_AUDIT.js`

Audits a nozzle candidate and offers promotion after passing.

> `RUN_MODEL_UPDATE.js`
>
> Updates debris/taxonomy datasets, trains candidates, and writes history.

> `RUN_REVIEW_MODEL_PROMOTION.js`
>
> Promotes selected completed candidates without retraining.

> **Do not launch numbered or lower-level Python scripts routinely** unless instructed by a maintainer or recovery procedure.

13. Outputs and Records

Location under `~/.openpnp2`	**Contents**
`scans/scan_*/`	Frames, detections, QA, logs, attempts, status, summaries.
`scans/manual_plate_*/`	Manual HiRes captures and run record.
`Plate_spreadsheets/`	CSV and formatted XLSX plate sheets.
`Plate_insect_images/P-<plate>/`	Canonical images by plate/well.
`Plate_audits/`	Completed-plate audit reports.
`Plate_taxonomy/`	Taxonomy outputs/review records.
`control/`	Status, flags, calibration, previews, process logs.
`scripts/BugPicker/Data/`	Models, reports, labels, feedback datasets.

Do not delete runs until spreadsheets, images, and taxonomy are verified and backed up.

14. Troubleshooting

Symptom	Response
HiRes gray, frozen, red-X, or absent	Halt. Close OpenPnP, reconnect Teslong USB, restart, home, and verify capture.
`invalid port descriptor: /dev/ttyACM0`	Restore/check the controller's Linux device, then restart OpenPnP.
Soft-limit error	Stop. Verify homing, offsets, selected slot, and calibration; do not simply raise limits.
N1 misses in X/Y	Stop; recheck tray seating, scan calibration, transform, and first-target position.
N1 too high or compressing specimens	Verify tray profile, seating/tilt, size class, and adaptive generation. Never exceed the physical floor.
Empty N1 reads occupied	Preserve images; label matching failures/empty captures, retrain, audit, then promote.
Training appears stuck	Check live log and CPU. If OpenPnP is closed, terminate only confirmed orphaned BugPicker workers.
Auto re-plate stops	Check open wells, unattempted targets, no-progress guard, and generation limit.
Images/rows missing	Preserve the scan and use audit/repair with a maintainer. Do not rename source files.
Script missing	Open `BugPicker → 00_Run_BugPicker`; refresh the menu or restart OpenPnP.

> After a collision, coordinate anomaly, or unexpected Z movement, identify the cause before retrying.

15. Shutdown and Handoff

☐ Confirm no run, taxonomy, update, or training process is active.

☐ Park safely and remove specimens/plates.

☐ Verify CSV/XLSX, images, audit, and taxonomy outputs.

☐ Record unresolved failures or hardware concerns.

☐ Close OpenPnP normally and power down per lab procedure.

16. Maintaining This Guide

Edit `docs/user-guide/source.html`, keep stable image filenames, and rebuild:

```
python3 docs/user-guide/build_user_guide.py
```

The builder ensures every current launcher is documented. Missing images become labeled placeholders. See `docs/user-guide/assets/README.md`.

☐ Update affected procedures and program entries.

☐ Replace screenshots when labels/layout change.

☐ Update safety language when motion changes.

☐ Build and resolve validation failures.

☐ Review HTML and PDF before release.
